



Contents lists available at ScienceDirect

Materials Today: Proceedings

journal homepage: www.elsevier.com/locate/matpr

A hybrid population based incremental learning algorithm with Particle Swarm Optimization for general threshold Visual Cryptography schemes

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ARTICLE INFO

Article history:

Received 16 December 2020

Received in revised form 16 December 2020

Accepted 28 December 2020

Available online xxxx

Keywords:

Visual cryptography (VC)
Particle Swarm Optimization (PSO)
Population-Based Incremental Learning (PBIL) algorithm
Hybrid population-based incremental learning algorithm with particle swarm optimization

ABSTRACT

An effortless transfer of multimedia information over the Internet is achieved due to advances in the network's technology. Commercial identifications, similarity maps, and numerous confidential data can be transferred over the Internet. When dealing with private videos and images, security challenges must be accounted for as hackers steal information by exploiting the weak links across communication networks. There has been the development of numerous schemes for the management of the security challenges of private images and videos. The encryption of a secret image onto n shares, wherein all participants can hold one or more shares, is known as the Visual Cryptography (VC) scheme. In the probabilistic VC scheme, the secret black pixels are reconstructed incompletely thus, the display quality obtained is not consistent with human visual perception. To enhance the display quality, it is required to optimize the threshold VCS. In this work, meta heuristic algorithms, Particle Swarm Optimization (PSO), and Population-Based Incremental Learning (PBIL) are modified to optimize the VCS threshold. PSO's abilities like simplicity, convergence speed, and optimum global searches, have made it the most frequently-employed evolutionary variant among hybrid techniques. Substitution of a population's individuals with a set of their statistics forms the basis of the concept behind the PBIL algorithm. Proposal for a hybrid Population-Based Incremental Learning algorithm with Particle Swarm Optimization (PBIL-PSO) has been provided in this work.

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1. Introduction

Of late, information is considered to be more valuable when it is shared with other people. The Internet has made it very easy to share information such as images, videos, and audio. However, there are security-related issues, especially since unauthorized data can be accessed by hackers. For the resolution of such issues, numerous techniques can be utilized. In today's world, the sharing of secret visual images like photographs, handwritten documentation, and several other types [1] within a computer-aided environment is also a critical issue.

The study of techniques for secure communication as well as practice is referred to as cryptography. It can also be defined as the art of transforming the information's plain form into its cipher form. Private key cryptography (symmetric) and public-key cryptography (asymmetric) are the two distinct types of cryptography.

A shared secret key is utilized in private key cryptography for carrying out encryption as well as decryption. On the other hand, different keys are used in the asymmetric key cryptosystem; while decryption uses the Private Key, encryption [2] will utilize the Public Key.

One of the latest image encryption schemes is Visual Cryptography [3], wherein the visual information gets encrypted to produce two or more shares that, although they do not have any individual information, can disclose the secret upon superposition. This scheme's key benefit pertains to the decoding procedure, not requiring any cryptographic knowledge and the human visual system's ability to decrypt secret messages. A novel encryption technique that is based on Particle Swarm Optimization has been presented in this work. This technique has been employed to get basis matrices. Afterward, encryption is done with these matrices.

As a representation of the perfect secrecy (with a one-time pad's utilization) concept as well as an extremely simple decrypt-

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<https://doi.org/10.1016/j.matpr.2020.12.1127>

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ing/decoding mechanism for secrets, Visual Cryptography is quite advantageous. Its most fascinating feature is that it is absolutely secure. From one-time padding to visual cryptography, there is a simplistic correlation. This can be considered as Visual Cryptography's second key benefit as other well-known existing cryptographic schemes are often only conditionally secure.

A family of optimization algorithms, designed for the resolution of mathematical problems' general classes through research procedures' combination to swiftly find the best solution, is referred to as metaheuristics. One such example is the Tabu search. The Tabu search will begin an initial solution and then try to enhance it through iterative transformation. For every iteration, there is a generation of the current solution's neighbourhood and also a selection of this neighbourhood's best solution. For the avoidance of circle, there a Tabu list's definition so as to halt any revisits to already assessed [4] solutions.

The rest of this investigation has been arranged as follows. The related literary works have been elaborated in Section Two. The various applied methodologies have been detailed in Section Three. While the experimentations' outcomes have been analyzed in Section Four, this investigation's conclusions have been provided in Section Five.

2. Literature review

There is a proposal for a secure image transmission scheme for the encrypted image's physical transmission with the Visual Cryptography Scheme (VCS). At the time of physical transmission, there a risk that inquisitive hackers may get attracted by meaningless shares. In the case of these shares' capture and stacking, there may be the secret's revelation. Yet another concern with regards to the VCS's physical implementation pertains to the transmission overhead increase as a result of the resultant multiple share images from one individual secret image's encryption. With a focus on both above-mentioned concerns, prior to the shares' transmission, there is the utilization of vector quantization (VQ) for each share's encoding as well as compression. For the VQ's utilization, its cell width and grid dimension parameters must be optimized for the diverse image types without any compromise on the shares' attribute of randomness. Therefore, Das et al., [5] had proposed a Particle Swarm Optimization-guided VQ. In addition to that, for the automatic attainment of each image type's optimal VQ at the time of transmission, this proposed VQ is trained in synchronism with the conjunction of a multilayer perceptron with an autoencoder. There is the successful deployment of the proposed scheme with diverse image types for secure physical transmission having 98.07% retrieval of image quality as well as 62.8% minimization of data volume.

The proposed visual cryptographic scheme in Geetha et al., [6] had three distinct phases: a) separation of color bands; b) generation of numerous shares; and c) optimal encryption and decryption. An Oppositional Gray Wolf Optimization (OGWO) based Elliptic Curve Cryptographic (ECC) technique had been suggested for the image's optimal encryption and decryption. At first, there is the input color image's separation into the three colour bands: R, G, and B. Afterwards, depending on the pixel measures, there is the generation of various image shares. Later, there is the partition of the image's shares into blocks as well as the original image's decryption with the OGWO algorithm's generated optimal key. It is evident from the experimental analysis outcomes that, without any deviations in the original image, the decrypted image had accomplished a PSNR of 59.012, an MSE of 0.11, and a CC of 1.

A novel visual cryptography scheme for a grayscale image that used Particle Swarm Optimization (PSO) had been propounded in Das et al., [7]. There was a gradual generation of the shares in this

proposed scheme. This was accomplished via a number of iterations from the secret input image through comparison of the image with the resultant image. The proposed scheme's outcomes demonstrated the stochastic optimization algorithm's strength in the visual cryptography field.

For enhancement of the conventional VCS's share management, the Tagged Visual Cryptography Scheme (TVCS) 1 will append tag images to the noise-like shares which are produced by the conventional VCS. Existing TVCSs may suffer from degradation in the recovered secret image's visual quality and may experience pixel expansion. For the resolution of these issues, Chiu & Lee [8] had devised a Threshold Adaptive Tagged Visual Cryptography Scheme (ATVCS). A mathematical optimization model is utilized for the ATVCS encryption problem's formulation. Later, there is the development of an evolutionary algorithm for the detection of this problem's optimal solution. With this proposed scheme, the encryptor will be able to tune the parameters for adjustment of the visual quality between the secret image and the tag image. This study's effectiveness and correctness are evident from the simulation outcomes.

Kaddouri & Omary [4] had given the design and implementation of asymmetrical encryption algorithm with an internal structure that was primarily based on the Tabu search algorithm. With this proposed heuristic, multiple searches could be done among different solutions, and the best solutions could be stored within an adaptive memory. Initially, a scheduling problem simulation is done to code the encryption problem. Later, an appropriate coding must be utilized for the proposed problem. This is followed by the utilization of a feasible evaluation function. The principles of encryption and decryption are illustrated by the proposed algorithm's generated symmetric key. Based on experiments conducted on this proposed algorithm, the end of this work has a description of its strengths as well as the elements for further improvement.

In Das et al., [9], a hybrid PSO-GA approach was employed for image encryption. The generated key was dependent on the Genetic Algorithm (GA) as well as the Particle Swarm Optimization (PSO). The different numbers are produced by the GA, and the key set's elements are chosen on the basis of the PSO. The logical operation between the image and the key-value has been utilized in the encrypted image's construction. Various applied tests between the plain image and the cipher image were conducted for verification of the proposed approach's effectiveness. Eventually, for almost every test, this approach has been found to provide better outcomes.

3. Methodology

The decryption procedure will directly stack the shared images, which are printed on transparencies. The objective is to maximize contrast under the blackness to obtain the best display quality of the image. Discussions related to the Tabu Search, the Particle Swarm Optimization, the Population-Based Incremental Learning algorithm, the Proposed Population-Based Incremental Learning-Tabu Search (PBIL-TS) Algorithm, and the Proposed Population-Based Incremental Learning-Particle Swarm Optimization (PBIL-PSO) have been provided in this section.

3.1. Tabu search (TS) algorithm

The Tabu search proposal was given by F.Glover in 1986. The algorithm is said to be Tabu as there is a prohibition on resuming searches from recently visited solutions [8]. The Tabu search algorithm is quite well-known due to its successful resolution of various problems. To the search space exploration [10] strategy, this algorithm has introduced a notion of memory. The Tabu search will

employ local or neighbourhood iterative procedures to shift from solution x towards a solution x' (within the vicinity of x) till the fulfillment of the stopping conditions.

The Tabu search's operating principle is dependent on a scheme to travel along with a solution space whilst continually seeking to enhance the current best solution, and by storing a list of the earlier moves in its memory, and therefore, guiding the research outside the zones which were earlier traversed. Research techniques applied in artificial intelligence were the inspiration for the Tabu search's fundamental concept. That is, to maintain a track of the research procedure's earlier path in one or more memories and to utilize this information for the future development's orientation. In effect, although it will not memorize all the displacements (extremely costly in memory), it does prevent access to certain solutions during a specific number of iterations. Definition of a solution's neighbourhood is given as an elementary transformation (movement) that will permit the switch from a solution to another nearby solution with a slight modification of the solution's structure. The basis for the Tabu search involves the below points:

- Utilization of flexible memory structures (short-term, medium-term as well as long-term) to facilitate complete exploration of the search history and the evaluation criteria.
- A control mechanism that is dependent on the alternation between the search procedure's liberating conditions (aspiration criterion) and the search procedure is constraining conditions (restriction Tabu).
- A merger of the search procedure's intensification strategy and diversification strategy:
 - o Intensification strategy will utilize the medium-term memory and will also aid in strengthening the search procedure in the recently-found best solutions' regions.
 - o The diversification strategy will utilize long-term memory and will also aid in searching the new regions

3.2. Particle Swarm Optimization (PSO)

Particle Swarm Optimization (PSO) is a stochastic, population-based evolutionary algorithm for the resolution of problems. As a type of swarm intelligence which is dependent on socio-psychological principles, the PSO will provide social, behavioral insights and will also contribute to applications in engineering. James Kennedy and Russell C. Eberhart [11] had originally devised the PSO in 1995. The PSO computational algorithm's population dynamics will mimic bio-inspired behaviour, that is, the behaviour of a "flock of birds". This will involve information sharing and will enable the particles to profit from the discoveries and the previous experiences of all other particles during the food search. In the PSO, every particle will have a randomized velocity vector (v) associated with it that traverses through the search space. Furthermore, every particle will track its solution space coordinates, that are associated with the best solution (fitness) accomplished by it so far, referred to as p (personal best position vector). PSO's global version will also track its overall best value (fitness). Its location, referred to as g (global best position vector), is attained from amongst the population's every particle. The particle's past best position, as well as the best overall position in the whole swarm, are utilized in order to attain the particle's new position vector (s) so as to minimize (or maximize) the fitness. At every time step t , there will be an update of the particle's velocity, and the particle will shift towards a new position.

A particle will shift towards an optimum solution [12] by utilizing the best position which has been explored by it and its neighbours. In accordance with a pre-defined fitness function, there is an assessment of every particle's performance (that is, a particle's

"closeness" to the global minimum). Consider the search space to be D -dimensional with m particles in the swarm.

Every particle will be situated at the position $X_i = [x_{i1}, x_{i2}, \dots, x_{iD}]$ with velocity $V_i = [v_{i1}, v_{i2}, \dots, v_{iD}]$, in which $i = 1, 2, \dots, m$. Each particle in the PSO algorithm will shift towards its own best position (p_{best}), indicated as $p_{best_i} = [p_{best_{i1}}, p_{best_{i2}}, \dots, p_{best_{iD}}]$ and the entire swarm's best position (g_{best}), indicated as $G_{best} = [g_{best_1}, g_{best_2}, \dots, g_{best_D}]$. Furthermore, all particles will modify its position in accordance with its velocity, that is randomly generated towards the p_{best} position and the g_{best} position. As per Eq. (1), expressions of the new velocity v_{is} and new position x_{is} for each particle i and dimension s are provided below:

$$v_{is}^t = wv_{is}^{t-1} + c_1b_1(p_{best_{is}}^{t-1} - x_{is}^{t-1}) + c_2b_2(g_{best_s}^{t-1} - x_{is}^{t-1})$$

$$x_{is}^t = x_{is}^{t-1} + v_{is}^t \quad (1)$$

wherein the iteration number is denoted as t . The inertial weight is denoted as w and is utilized for controlling the velocity as well as the balance of the algorithm's an exploration and exploitation capabilities. When the value of w is large, the particles will be at high velocity and will be prevented from getting captured in the local optima. On the other hand, when the value of w is small, the particles will be at low velocity and will be encouraged to exploit the same area of search. The acceleration coefficients, constants c_1 and c_2 , will determine the particles' preference to shift nearer towards the p_{best} position or the g_{best} position. Two independent random numbers that are uniformly distributed between 0 and 1 are denoted as b_1 and b_2 . The maximum number of generations, the designated value of p_{best} , or no further improvement in p_{best} will constitute the PSO algorithm's termination criterion. Ease of implementation, tuning with fewer parameters, and avoidance of the local minima [12] are some of the PSO's attributes.

Each time step of the PSO concept will comprise of the velocity change of each particle traversing towards its personal best location as well as its global best location with an inertia weight ω , that will control the earlier velocity's magnitude in the new velocity's assessment. There is the weighting of the velocity by random terms; wherein there is the generation of separate random numbers (r_1 and r_2) respectively for velocities towards the personal best location and the global best location. For the optimization technique's purpose of learning, the social parameters c_1 and c_2 are utilized.

3.3. Population-Based Incremental Learning (PBIL)

Originally proposed by Baluja and Caruana (1995), the Population-Based Incremental Learning (PBIL) algorithm is an evolutionary optimization algorithm as well as an estimation of distribution algorithm. This algorithm is part of a family of metaheuristics, which is utilized for the detection of near-optimal solutions to complex problems which are often, at least NP -hard. The implication is that these problems will usually have large and discrete solution spaces wherein the problem's resolution time will grow as a polynomial with the size of the problem (that is, the number of variables). The PBIL is a kind of Genetic Algorithm (GA) in which there is explicit maintenance of the statistics enclosed within any population; there is the evolution of the whole population's genotype instead of the evolution of the individual members' genotypes.

Below is the PBIL algorithm:

- (1) Generation of a population from a probability vector,

Table 1

Contrast for Proposed PBIL-PSO.

(k, n) for different Blackness Range (100,90)	Contrast % - Tabu	Contrast %-PSO	Contrast %-PBIL	Contrast %-PBIL-Tabu	Contrast %-PBIL-PSO
(2,3)	34.2	35.1	36	37	37.9
(2,4)	30.1	30.9	31.7	32.5	33.5
(2,5)	29.4	30.2	31	31.8	32.7
(2,6)	28.6	29.2	30	30.8	31.4
(2,7)	26.4	27.2	27.9	28.6	29.2
(2,8)	26.2	26.8	27.4	28.1	28.7

Table 2

Contrast for Proposed PBIL-PSO.

(k, n) for different Blackness Range (90,80)	Contrast % - Tabu	Contrast %-PSO	Contrast %-PBIL	Contrast %-PBIL-Tabu	Contrast %-PBIL-PSO
(2,3)	35.4	36.3	37.1	38.1	39.1
(2,4)	31.2	32	32.8	33.7	34.4
(2,5)	30.4	31.3	32.2	33	33.7
(2,6)	29.6	30.4	31.2	32	32.7
(2,7)	27.4	28.1	28.8	29.4	30.2
(2,8)	27.2	27.8	28.6	29.4	30.2

Table 3

Contrast for Proposed PBIL-PSO.

(k, n) for different Blackness Range (80,70)	Contrast % - Tabu	Contrast %-PSO	Contrast %-PBIL	Contrast %-PBIL-Tabu	Contrast %-PBIL-PSO
(2,3)	36.6	37.5	38.4	39.5	40.3
(2,4)	32.3	33.1	34.1	34.8	35.8
(2,5)	31.3	32.1	32.8	33.7	34.4
(2,6)	30.6	31.3	32.1	32.9	33.8
(2,7)	28.2	28.9	29.7	30.4	31.2
(2,8)	28	28.7	29.5	30.2	31

Table 4

Contrast for Proposed PBIL-PSO.

(k = 5, n) for different Blackness Range (100,90)	Contrast % - Tabu	Contrast %-PSO	Contrast %-PBIL	Contrast %-PBIL-Tabu	Contrast %-PBIL-PSO
(2,5)	29.4	30.2	31	31.8	32.7
(3,5)	19.8	20.2	21.6	21.3	21.5
(4,5)	13.2	14.2	14.3	14.8	14.5

Table 5

Contrast for Proposed PBIL-PSO.

(k = 5, n) for different Blackness Range (90,80)	Contrast % - Tabu	Contrast %-PSO	Contrast %-PBIL	Contrast %-PBIL-Tabu	Contrast %-PBIL-PSO
(2,5)	30.4	31.3	32.2	33	33.7
(3,5)	21.2	21.6	21.6	23.5	22.5
(4,5)	15.1	15	14.2	16.1	15.6

Table 6

Contrast for Proposed PBIL-PSO.

(k = 5, n) for different Blackness Range (80,70)	Contrast % - Tabu	Contrast %-PSO	Contrast %-PBIL	Contrast %-PBIL-Tabu	Contrast %-PBIL-PSO
(2,5)	31.3	32.1	32.8	33.7	34.4
(3,5)	21.4	21.5	22.9	24.1	24.4
(4,5)	14.3	14.8	15.7	16.8	17.1

- (2) Every member will be assessed and ranked depending on their individual fitness,
- (3) Depending on the fittest individuals, there is an update of the population probability vector,
- (4) Mutation,
- (5) Repetition of Step 1 to Step 4.

In comparison to typical GAs, PBIL [13] has had much success on multiple benchmarks as well as real-time problems like power system stabilizer advancement with incremental learning. With

incremental learning's application, training can be performed. Later on, the pattern variations could be utilized for developing models for responsive stabilizer designs.

3.4. Proposed Population-Based Incremental Learning-Tabu search (PBIL-TS) algorithm

PBIL algorithm will obtain a probability model which is based on the current excellent individuals' learning, and also will utilize probability model control to generate a new group. The lack of uti-

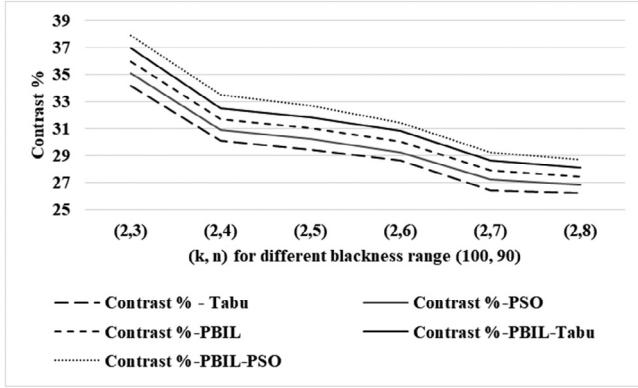


Fig. 1. Contrast for Proposed PBIL-PSO.

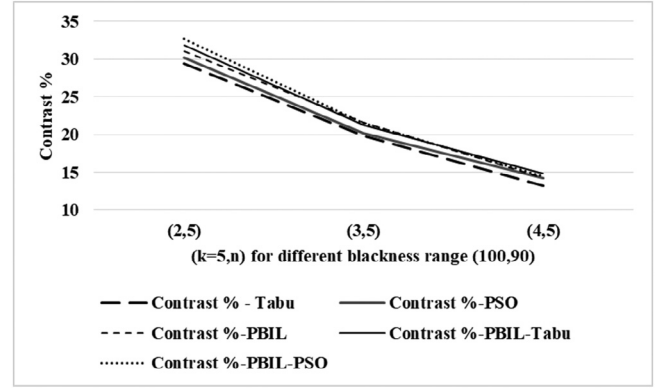


Fig. 4. Contrast for Proposed PBIL-PSO.

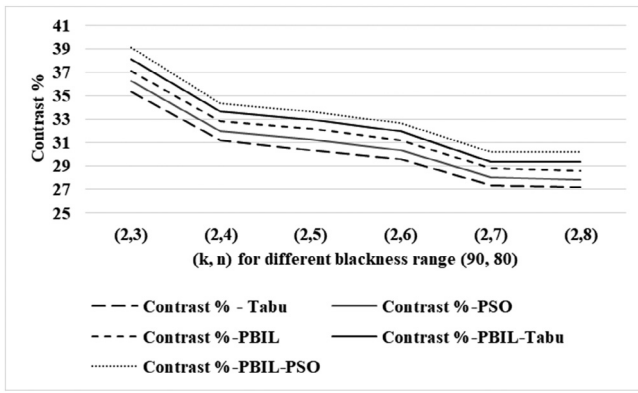


Fig. 2. Contrast for Proposed PBIL-PSO.

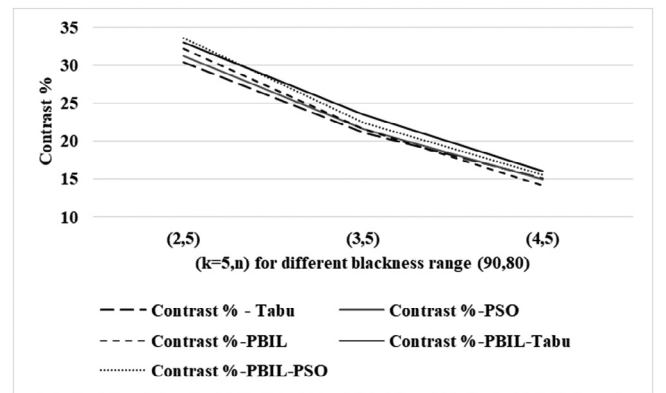


Fig. 5. Contrast for Proposed PBIL-PSO.

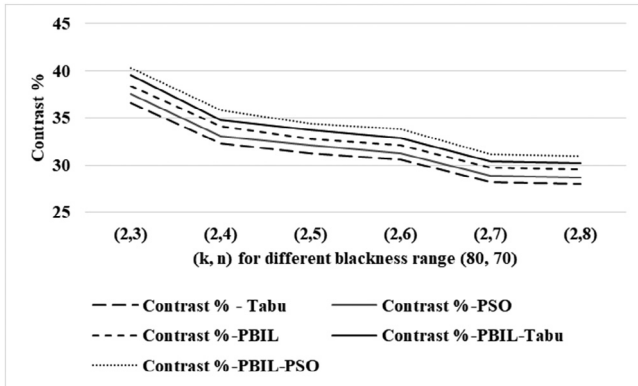


Fig. 3. Contrast for Proposed PBIL-PSO.

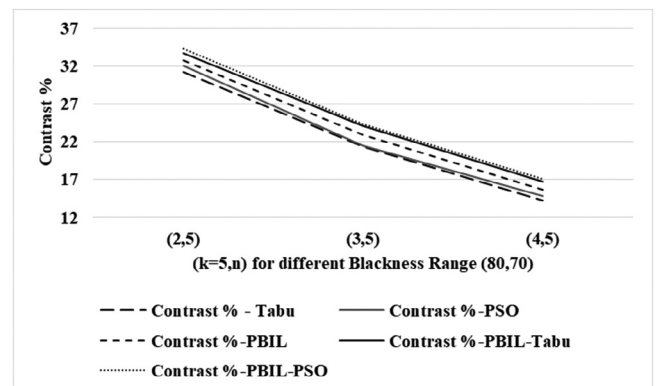


Fig. 6. Contrast for Proposed PBIL-PSO.

lization of the genetic operator will result in the avoidance of the deception problem as well as and the conventional genetic algorithm's chain problem. This, in turn, will prevent the algorithm's fall into either premature convergence or local optimization. Despite the PBIL's recent introduction, it is now primarily employed for the resolution of single-objective optimization problems in combinatorial optimization [14], production scheduling, aviation military, engineering optimization, and biological information.

Below is the Tabu search with the PBIL algorithm:

1) Obtain the population's initial solution (initialization),

- 2) Produce a list of the candidates' movements. In addition to that, assess every members' fitness functions.
 - 3) Pick the best candidate based on the Tabu restrictions as well as the aspiration criteria. The choice will be an alternative that would not be registered only when it is better than the previous solution.
 - 4) Perform the mutation procedure.
 - 5) Application of the stopping criterion.
- Continue: Modify the candidates of eligibility. Shift to 2
 - Stop: Shift towards the intensification and diversification strategies.

3.5. Proposed population-based incremental learning-particle Swarm Optimization (PBIL-PSO)

Nowadays, for the resolution of optimization problems in engineering as well as other fields, the Population-Based Incremental Learning (PBIL) algorithm has been utilized. Below is the Hybrid Particle Swarm Optimization-Population based Incremental Learning's algorithm:

- Consider the Particle Swarm Optimization's initial population. The probability vector is used for the population's generation.
- Creation of a vector having length that is equivalent to the required chromosome. Moreover, this vector's elements will have 1-valued probabilities in the chromosome's respective bit position. There is the vector's initialization of 0.5 in all bits when the chromosomes employ a binary alphabet.
- Generation of a number of samples from the vector in which the current probability vector will determine the probability of a 1 in each bit position of each vector.
- There is an association of the largest function values from each temperature reduction's end with the maximum current function value as well as the optimum function value to bring the algorithm to a halt. When all these changes are lesser in comparison to a pre-definite small value, there is the algorithm's termination.
- The procedure of mutation.
- From this population, there is the identification of the fittest chromosome(s).
- Modification of the probability vector's elements such that the positions will have an increase in 1-valued probabilities wherever the fittest chromosomes are 1.

In this procedure, there is the initialization of all elements of the probability as 0.5. When any of these elements approach either 0 or 1, the procedure will be halted. A supervised learning technique is employed for updating the probability vector's elements.

4. Results and discussions

In this section, the Tabu, PSO, PBIL methods, hybrid PBIL-Tabu, and hybrid PBIL-PSO are used. Experiments are carried out using $(k, n) = (5, n)$ for different blackness range. The contrast (various blackness ranges) as shown in Tables 1 to 6 and Figs. 1 to 6.

From Fig. 1, it can be observed that the PBIL-PSO has higher contrast by 10.3%, by 7.7%, by 5.14% and by 2.4% for (2, 3), by 10.63%, by 7.95%, by 5.34%, and by 2.79% for (2, 5), by 10.1%, by 7.1%, by 4.6%, and by 2.1% for (2, 7), and by 9.12%, by 6.85%, by 4.63% and by 2.11% for (2, 8) respectively for Tabu, PSO, PBIL, and PBIL-Tabu in blackness range (100, 90).

From Fig. 2, it can be observed that the PBIL-PSO has higher contrast by 9.93%, by 7.43%, by 5.25% and by 2.6% for (2, 3), by 10.3%, by 7.4%, by 4.6% and by 2.1% for (2, 5), by 9.72%, by 7.2%, by 4.75%, and by 2.7% for (2, 7), and by 10.5%, by 8.3%, by 5.44% and by 2.7% for (2, 8) respectively for Tabu, PSO, PBIL, and PBIL-Tabu in blackness range (90, 80).

From Fig. 3, it can be observed that the PBIL-PSO has higher contrast by 9.62%, by 7.2%, by 4.83% and by 2% for (2, 3), by 9.44%, by 6.92%, by 4.76% and by 2.1% for (2, 5), by 10.1%, by 7.65%, by 4.93%, and by 2.59% for (2, 7), and by 10.2%, by 7.71%, by 4.95% and by 2.61% for (2, 8) respectively for Tabu, PSO, PBIL, and PBIL-Tabu in blackness range (80, 70).

From Fig. 4, it can be observed that the PBIL-PSO has higher contrast by 10.63%, by 7.95%, by 5.34% and by 2.79% for (2, 5), by 8.23%, by 6.24%, by 0.5%, and by 0.93% for (3, 5), and by 9.39%, by 2.1%, by

1.39% and by 2.05% for (4, 5), respectively for Tabu, PSO, PBIL, and PBIL-Tabu in blackness range (100, 90).

From Fig. 5, it can be observed that the PBIL-PSO has higher contrast by 10.3%, by 7.4%, by 4.6% and by 2.1% for (2, 5), by 5.95%, by 4.1%, by 4.1%, and by 4.35% for (3, 5), and by 3.26%, by 3.92%, by 9.4% and by 3.15% for (4, 5), respectively for Tabu, PSO, PBIL, and PBIL-Tabu in blackness range (90, 80).

From Fig. 6, it can be observed that the PBIL-PSO has higher contrast by 9.44%, by 6.92%, by 4.8% and by 2.1% for (2, 5), by 13.1%, by 12.64%, by 6.34%, and by 1.24% for (3, 5), and by 17.83%, by 14.42%, by 8.54% and by 1.77% for (4, 5), respectively for Tabu, PSO, PBIL, and PBIL-Tabu in blackness range (80, 70).

5. Conclusion

Visual Cryptography will facilitate the transfer of visual information and will also encompass this domain's various aspects, ranging from its origin to the latest schemes that are being applied as well as actively researched. Being a simple and effective scheme, Visual Cryptography (VC) is capable of encrypting digital images without any complex computations. This scheme will encrypt a secret image onto a number of shares, wherein none of the individual shares have the ability to reveal this secret image. Decoding of the secret image with the human visual system is done through the stacking of every share. Thus, the VC scheme has ease of implementation and is also more secure. Experimental outcomes have demonstrated that the PBIL-PSO has higher contrast by 10.3%, by 7.7%, by 5.14%, and by 2.4% for (2, 3), by 10.63%, by 7.95%, by 5.34%, and by 2.79% for (2, 5), by 10.1%, by 7.1%, by 4.6%, and by 2.1% for (2, 7), and by 9.12%, by 6.85%, by 4.63%, and by 2.11% for (2, 8), respectively for Tabu, PSO, PBIL, and PBIL-Tabu in blackness range (100, 90).

CRedit authorship contribution statement

Kanusu Srinivasa Rao: Conceptualization, Methodology, Software, Visualization, Writing - original draft. **Sridhar Mandapati:** Data curation, Supervision, Validation.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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